THE TOTAL TENSOR PRODUCT OF TWO K-FLAT COMPLEXES IS K-FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- K-flat complex, defined as a complex whose tensor product functor preserves quasi-isomorphisms. definition environment at source lines 14185-14191 of the same chapter. Supplies the property to be proved for Tot(K tensor L) and the properties assumed for K and L.
- Associativity of the total tensor product of complexes, as displayed. the displayed isomorphism inside the proof, source lines 14231-14235. Rewrites a test against Tot(K tensor L) as tests against K and against L, which is the whole argument.
- Total complex of a tensor product of complexes and its differential. introduced in the same section, source lines 14000-14184. Makes sense of the two sides of the displayed isomorphism.

Source reference. Original lemma 0795: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9260–9260. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a ring. If $K^\bullet$, $L^\bullet$ are K-flat complexes of R-modules, then $Tot(K^\bullet \otimes_R L^\bullet)$ is a K-flat complex of R-modules.*

Proof. Follows from the isomorphism

$$\mathrm{Tot}(M^\bullet \otimes_R \mathrm{Tot}(K^\bullet \otimes_R L^\bullet)) = \mathrm{Tot}(\mathrm{Tot}(M^\bullet \otimes_R K^\bullet) \otimes_R L^\bullet)$$

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and the definition. □